

Hedging Analysis with Greeks

Project Summary

This project studies delta-hedged option P&L under different volatility assumptions, with a focus on the distinction between actual volatility and implied volatility, the impact of Gamma, and the mechanics of Gamma scalping, regime shift involved. The analysis combines continuous-time Black-Scholes derivations with discrete-time Monte Carlo simulation.

The first part compares Euler-Maruyama, Milstein, and Sobol sequence with Brownian Bridge methods for simulating geometric Brownian motion. The results show that Euler and Milstein generate visually similar paths, but Milstein achieves better convergence accuracy, while Sobol with Brownian Bridge efficiently reduce the variance.

The second part analyzes hedging with actual volatility as a theoretical benchmark and hedging with implied volatility as a practical variance-spread framework. The results show that Gamma-scalping profitability is highly path-dependent: realized volatility need to exceed implied volatility, the spot price remains close to the strike, particular approaching to expiry, and Gamma is still elevated.

Overall, the project provides a structured bridge between Black-Scholes theory, numerical simulation, practical volatility-trading intuition, while highlighting the limitations of continuous-time assumptions and the importance of regime sensitivity.

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1. Geometric Brownian Motion Asset Simulation

Here it will involve three methods: Euler-Maruyama scheme, Milstein scheme, and Sobol sequence with the Brownian Bridge method.

As for the numerical approximations for stochastic differential equation (SDE), Euler-Maruyama and Milstein schemes are typical discretization methods to do simulations.

Starting with the general form of SDE:

$$dY_t = a(t, Y_t)dt + b(t, Y_t)dX_t$$

Where:

Y_t : stochastic process

$a(t, Y_t)$: drift term

$b(t, Y_t)$: diffusion term

X_t : standard Brownian motion (Wiener process)

1.1 Euler-Maruyama Scheme

It is a discrete approximation, Taylor expansion to first order. Given a time step δt , the simulation is

$$Y_{t+\delta t} = Y_t + a(t, Y_t)\delta t + b(t, Y_t)dX_t$$

where dX_t is the Brownian increment

$dX_t = X_{t_{i+1}} - X_{t_i}$, $dX_t \sim N(0, \delta t)$. It is typically simulated as: $dX_t = \sqrt{\delta t}\phi_i$, $\phi_i \sim N(0,1)$

Namely, simulation Brownian motion means drawing standard normal at each step.

For geometric Brownian motion, the SDE formula is $dS_t = \mu S_t dt + \sigma S_t dX_t$,

so the Euler scheme formula is $S_{t+\delta t} = S_t(1 + \mu\delta t + \sigma\sqrt{\delta t}\phi_i)$ (1)

1.2 Milstein Scheme

It includes a second-order term from the Itô -Taylor expansion to improve accuracy.

The formula is: $Y_{t+\delta t} = Y_t + a(t, Y_t)\delta t + b(t, Y_t)dX_t + \frac{1}{2}b(t, Y_t)\frac{\partial}{\partial Y}b(t, Y_t)((dX_t)^2 - \delta t)$

The extra term $\frac{1}{2}b(t, Y_t)\frac{\partial}{\partial Y}b(t, Y_t)((dX_t)^2 - \delta t)$ is the Itô correction, and it captures the curvature of the diffusion coefficient.

For geometric Brownian motion:

$$b(t, Y_t) = \sigma S_t,$$

$$\frac{\partial}{\partial Y} b(t, Y_t) = \frac{\partial}{\partial S_t} (\sigma S_t) = \sigma$$

So the Milstein scheme formula becomes:

$$S_{t+\delta t} = S_t(1 + \mu\delta t + \sigma\sqrt{\delta t}\phi_i + \frac{1}{2}\sigma^2(\delta t\phi_i^2 - \delta t)) \quad (2)$$

Milstein scheme may achieve better pathwise accuracy by capturing the second-order stochastic structure.

1.3 Properties

Euler-Maruyama Scheme:

It is of simple implementation, and minimal computational cost

It has strong order of convergence: $O(\sqrt{\delta t})$

It fails to capture curvature of diffusion term

It has weak order of convergence 1

Milstein scheme:

It requires computing derivative of the diffusion term

It has strong order of convergence $O(\delta t)$, twice as fast as Euler in terms of step-size reduction

It captures Itô-Taylor expansion up to $O(\delta t)$

It has weak order of convergence 1

Strong/weak convergence evaluates the error derived from the approximation of discrete-time (finite δt) paths to the true continuous-time process. Convergence order reflects how fast the error shrinks with δt . Both Euler and Milstein have weak order 1, but Milstein has strong order 1 while Euler's strong order is $\frac{1}{2}$, which matters for path-dependent payoffs.

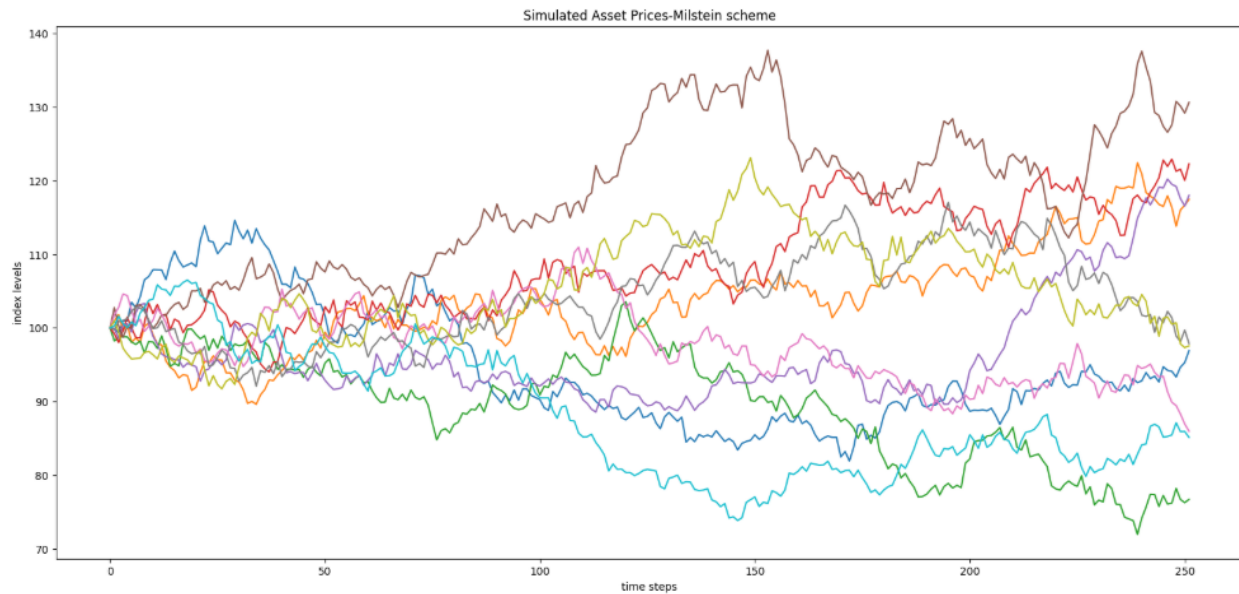
For European options (payoff only depends on terminal), weak convergence (distributional accuracy) suffices. Euler with $\delta t = 1/252$ (daily steps) is adequate.

For path-dependent options, strong convergence (pathwise/trajectory accuracy) matters more. Milstein is preferred. The discretization error of Euler may cause mispricing.

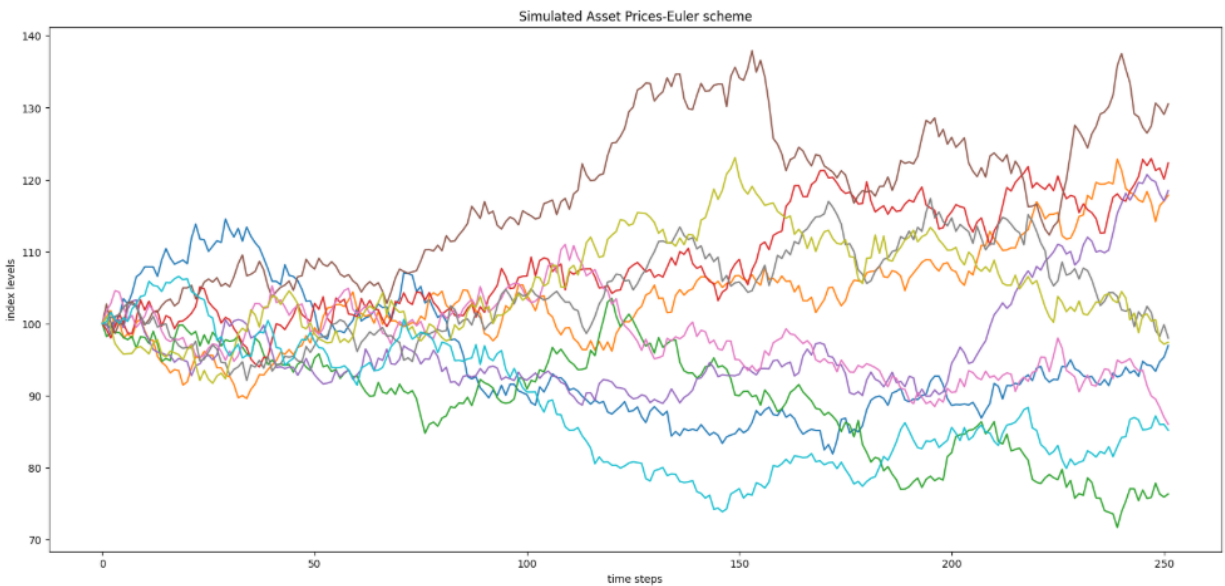
1.4 Path Simulation and Observations

Using formula (1) and (2) for implementation.

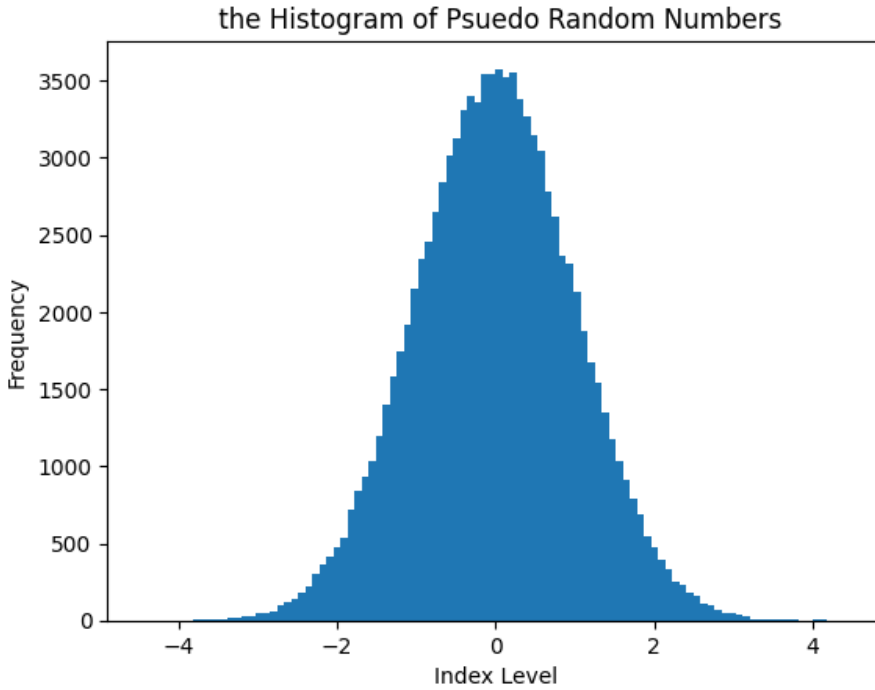
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Text(0.5, 1.0, 'Simulated Asset Prices-Milstein scheme')
```



```
Text(0.5, 1.0, 'Simulated Asset Prices-Euler scheme')
```

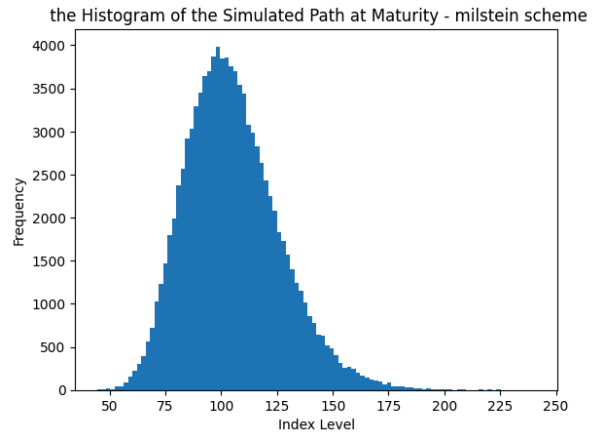
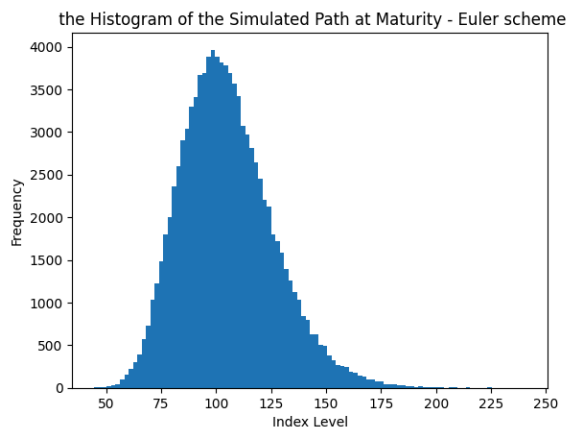


Scenario: $S_0=100$, rate=0.05, sigma= 0.2, time period=1, time steps=252



```
pseudorandom.mean(), pseudorandom.std()
```

```
(-0.005543773317243638, 1.0019601254713244)
```



Observations:

- The path simulations and terminal distributions of Euler and Milstein schemes do not show remarkable distinctions. Their difference is in accuracy, not visual appearance. Improvement of accuracy is numerically subtle. Based on formulae of Euler scheme and Milstein scheme, dominate noise term $\sigma dX_t \sim O(\sqrt{\delta t})$, and

Milstein correction term $\frac{1}{2}\sigma^2((dX_t)^2 - \delta t) \sim O(\delta t)$, with zero expectation,

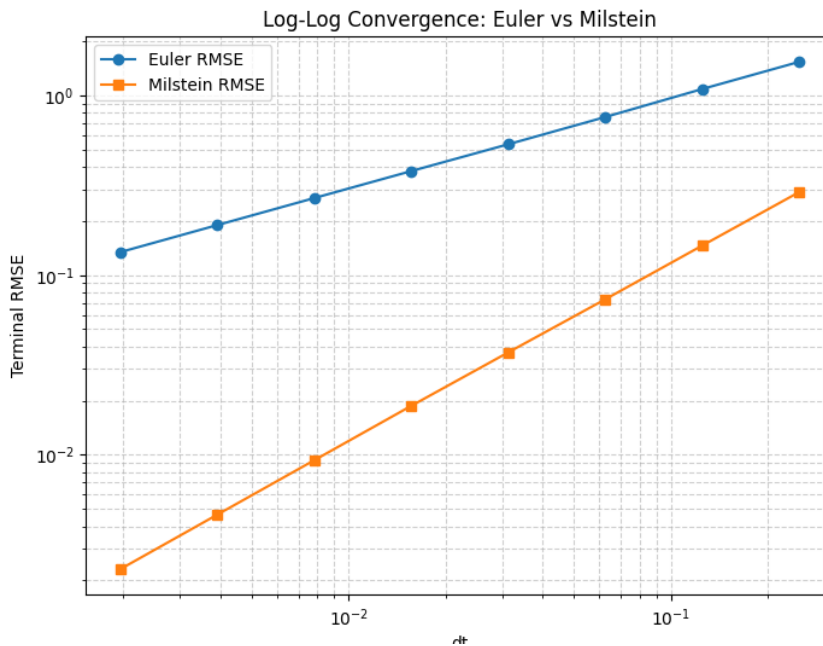
$\mathbb{E}[(dX_t)^2 - \delta t] = 0$. The leading random fluctuation hides the Milstein correction visually. Therefore, it is necessary to implement further error checking.

- b) The histogram of pseudo random numbers, and the statistical mean (-0.005544) and standard deviation (1.001960) show a standard normal distribution.
- c) Both histograms of simulated path at maturity indicate approximate lognormal, as expected for geometric Brownian motion.

1.5 Error Checking

Generally, there are two approaches for error checking, terminal-value error check and pathwise max error. Given more focus on the expectation of payoff, particular for Vanilla European option pricing, the terminal-value error checking suffices.

Here, root mean square values are calculated as error evaluation (RMSE), and time steps vary from 4 to 512. The plot is log-log form.



Estimated Euler slope: 0.5024190440759594
Estimated Milstein slope: 0.995268051466122

The plot and values show that Euler slope ≈ 0.5 and Milstein slope ≈ 1 . Euler error is larger than Milstein error. As dt decreases, the error gap between two schemes becomes wider.

1.6 Sobol Sequence with the Brownian Bridge Method

1.6.1 Sobol Sequence

Sobol sequence are low-discrepancy sequences. Unlike pseudo-random numbers, which mimic randomness, Sobol numbers are quasi-random, namely they follow a specific pattern to fill space more uniformly. Sobol sequences minimize discrepancy, avoiding clustering and gaps seen in pseudo-random numbers. They have better convergence properties. Its convergence rate is $O(\frac{(\log N)^d}{N})$, which is faster than $O(\frac{1}{\sqrt{N}})$ for pseudo-random.

It is built by using direction numbers and bitwise XOR operations on binary fractions.

Direction Numbers: $v_k^{(j)}$, where j is dimension, $j=1, 2, \dots, d$.

They are chosen based on primitive polynomials over $GF(2)$. Initial q direction numbers $v_1^{(j)}, v_2^{(j)}, \dots, v_q^{(j)}$ are freely chosen, subjected to be odd and less than 2^i . For $i > q$, direction numbers are generated using XOR operations and right shifts.

Generating the sequence: for the n -th point in dimension j , use the binary reflected Gray code to determine which direction numbers to XOR operation.

The result is a sequence of points $\{x_n = (x_n^{(1)}, x_n^{(2)}, \dots, x_n^{(d)})\}$ in $[0,1)^d$, where each coordinate is generated using above machinery.

Sobol sequence scramble the sequences and use Power-of-2 Sample Sizes: $N = 2^m$ for best results.

1.6.2 Brownian Bridge Method

The Brownian bridge is a continuous-time stochastic process derived from Brownian motion by conditioning on already computed points. Instead of simulating a Brownian motion (Wiener process) path incrementally and sequentially from start to end, the

Brownian Bridge constructs the path recursively and conditionally by filling in the intermediate values. It builds paths hierarchically.

Standard Brownian motion X_t at discrete time points $t_i = i\delta t$, where $\delta t = \frac{T}{n}$, $i = 0, 1,$

$2, \dots, n$

$$X_{t_{i+1}} = X_{t_i} + \sqrt{t_{i+1} - t_i} \cdot \phi_i, \text{ namely } dX_t = \sqrt{\delta t} \phi_i$$

Where $X_{t_0} = 0$

$\phi_i \sim N(0,1)$, are independent standard normal variates

The increments $X_{t_{i+1}} - X_{t_i}$ are normally distributed

Brownian bridge path construction

The Brownian bridge first fixes the final value, and then recursively fills in the intermediate points.

Final value: $X_T = \sqrt{T} \cdot \phi$, $\phi \sim N(0,1)$ (3)

Recursive refinement:

For any intermediate time t_j between t_i and t_k ($t_i < t_j < t_k$), the value of X_{t_j} is conditioned on X_{t_i} and X_{t_k} :

Conditional mean:

$$\mathbb{E}[X_{t_j} | X_{t_i}, X_{t_k}] = X_{t_i} + \frac{t_j - t_i}{t_k - t_i} (X_{t_k} - X_{t_i})$$

Conditional variance:

$$\mathbb{V}[X_{t_j} | X_{t_i}, X_{t_k}] = \frac{(t_j - t_i)(t_k - t_j)}{t_k - t_i}$$

$$X_{t_j} = \mathbb{E}[X_{t_j} | X_{t_i}, X_{t_k}] + \sqrt{\mathbb{V}[X_{t_j} | X_{t_i}, X_{t_k}]} \cdot \phi_j \quad (4)$$

where $\phi_j \sim N(0,1)$

This method assumes the underlying process is Gaussian. Its variance reduction is achieved by conditioning on already known points.

1.6.3 Brownian Bridge with Sobol Sequence

Brownian bridge works well with quasi-random numbers (e.g. Sobol). It reduces effective dimensionality by concentrating payoff-relevant information into early Sobol dimensions. The first few dimensions of a Sobol sequence are the most uniformly distributed, which is aligned with the most important features of the path and reduces the variance of the Monte Carlo estimator compared to a standard sequential random walk. Its effective reduction of effective dimension is crucial for high-dimensional path simulation.

1.6.4 Path Simulation

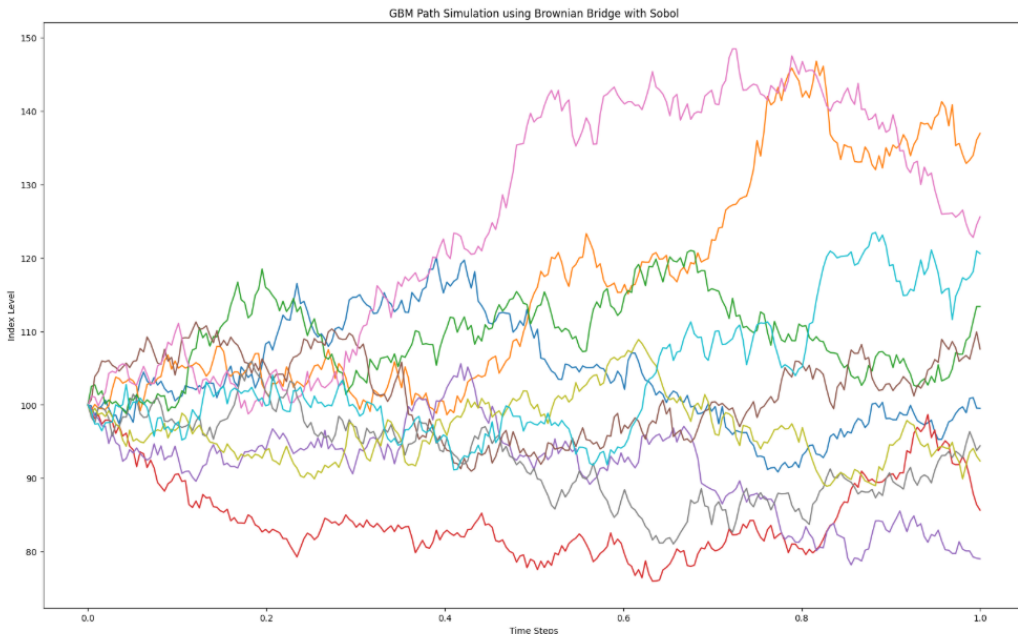
For implementations of Brownian bridge with Sobol, the iterative method is chosen, instead of recursive one, since iterative method is more robust and it shows the dyadic construction sequence more explicit.

It uses formula (3) to calculate terminal values, and uses formula (4) to calculate Brownian bridge process. ϕ_j in both formulae are generated by Sobol sequences.

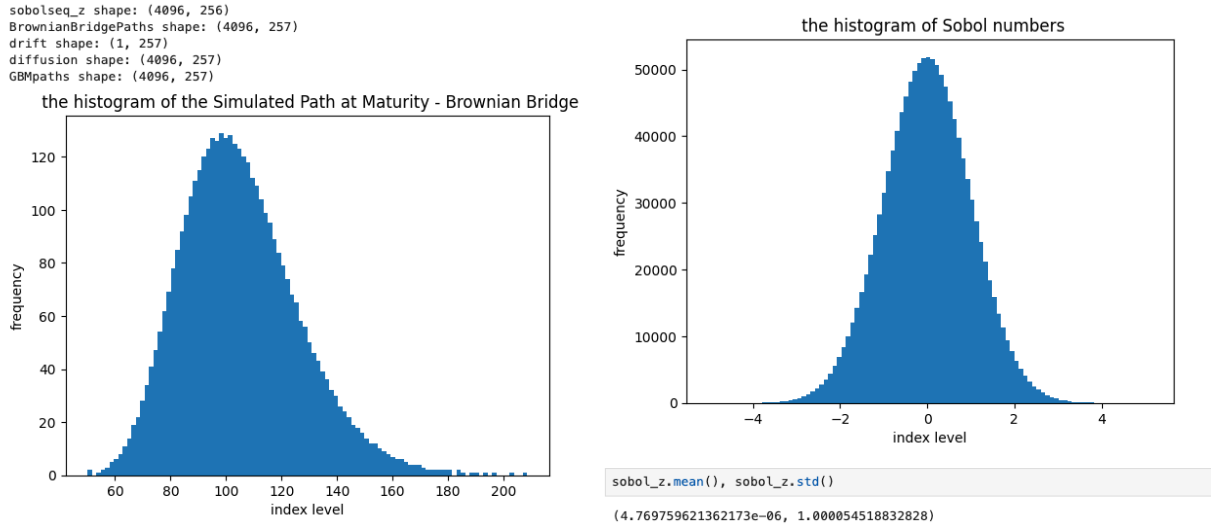
Based on geometric Brownian motion: $\frac{dS_t}{S_t} = \mu dt + \sigma dX_t$,

$$S_t = S_0 e^{(\mu - \frac{1}{2}\sigma^2)t + \sigma X_t} \quad (5)$$

It uses formula (5) to simulate path S_t , where X_t is the Brownian bridge process.



Scenario: $S_0=100$, rate=0.05, sigma= 0.2, time period=1, time steps=256



Observations:

- The simulated paths show continuous trajectories, with no visible artifacts. Dispersion increases over time.
- The histogram at maturity (terminal distribution) shows right-skewed with long right tail, consistent with the distribution of S_t .
- The Sobol sequence shows reasonable mean and standard deviation, mean ≈ 0 , variance ≈ 1 . The histogram displays symmetric and bell-shaped, reflecting the simulation efficiency of Sobol generator.

1.6.5 Quantitative Comparison of Variance Reduction

This subsection implements the RMSE benchmark to quantify the variance reduction achieved by Sobel sequence combined with the Brownian bridge.

$$RMSE = \sqrt{\frac{1}{R} \sum_{r=1}^R (\hat{V}_{MC,r} - V_{BS})^2}$$

where $\hat{V}_{MC,r}$ is the Monte Carlo price from replication r , and V_{BS} is the Black-Scholes analytical price. For each choice of sample size N , it runs R independent replications and compute RMSE across these replications to isolate sampling error.

Monte Carlo simulation:

$$\hat{V}_{MC} = e^{-r(T-0)} \frac{1}{N} \sum_{i=1}^N \max(S_T^{(i)} - E, 0)$$

where $S_T^{(i)}$ is simulated terminal price for path i . To focus purely on sampling efficiency, we feed the paths from exact GBM solution, rather than Euler or Milstein time discretization scheme.

$$S_T = S_0 e^{(\mu - \frac{1}{2}\sigma^2)(T-0) + \sigma(X_T - X_0)}$$

where $X_0 = 0$, with $X_t \sim N(0, T)$

Two simulations for X_t are:

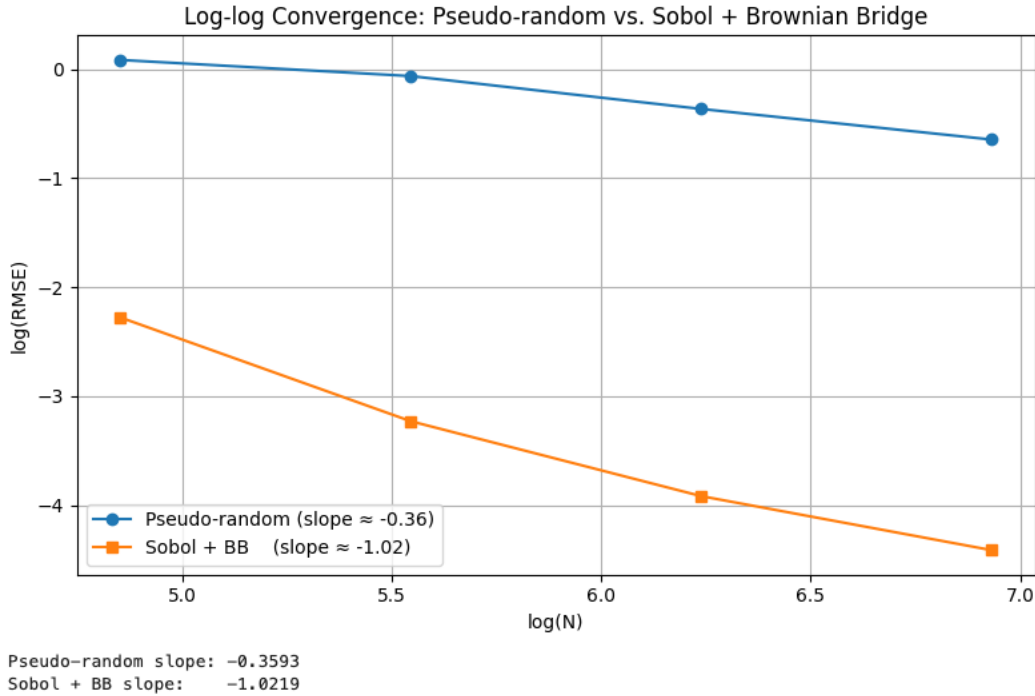
	Random number generator	Path construction	Brownian motion type
Simulation A	Pseudo-random (i.i.d. standard normals)	Sequential increments	Standard Brownian motion built forward step-by-step
Simulation B	Sobol sequence (low-discrepancy quasi-random)	Brownian Bridge (hierarchical, dyadic)	Brownian Bridge process conditioned on terminal value, filled recursively

In both cases, the option parameters, maturity, and number of paths N are identical.

The resulting RMSE values for different N are:

N	A(RMSE)	B(RMSE)
128	1.086664	0.102472
256	0.937399	0.039599
512	0.693482	0.019936
1024	0.523811	0.012149

The table shows that Simulation B consistently outperforming Simulation A by a wide margin at every N , typically delivering an order-of-magnitude smaller RMSE.



The log-log plot highlights the difference in convergence speed: that pseudo-random scheme exhibits a slope of approximately -0.36, while Sobol sequence plus Brownian Bridge has a slope near -1.02, indicating much faster quasi-Monte Carlo convergence in this parameter range.

2. Hedge with Actual Volatility & Hedge with Implied Volatility

This section demonstrates how to capture volatility risk premium through delta hedging, quantifying the gamma exposure that drives profitability.

2.1 Hedge with Actual Volatility

2.1.1 Mathematical Derivatives of the Known Total P&L

We need to construct a portfolio and calculate the mark-to market profits.

Assumption:

- the option is vanilla call option: $V_i(S, t; \sigma_i)$, $V_a(S, t; \sigma_a)$
- the underlying asset follows geometric Brownian motion: $dS = \mu S dt + \sigma_a S dX_t$
- assets follow risk-neutral drift: $\mu = r$
- no dividends $D=0$

the portfolio is set up by buying one unit of option V_i and short Δ_a units of stocks.

At time t , the hedging portfolio:

- a) option: V_i
- b) stocks: $-\Delta_a S$
- c) cash flows: $-V_i + \Delta_a S$

At time $t + dt$, the portfolio value becomes:

- a) option: $V_i + dV_i$
- b) stocks: $-\Delta_a S - \Delta_a dS$
- c) cash flows: $(-V_i + \Delta_a S)(1 + rdt)$

so, the mark-to-market profit over time dt :

$$dV_i - \Delta_a dS + (-V_i + \Delta_a S)r dt$$

$$= dV_i - \Delta_a dS - r(V_i - \Delta_a S)dt \quad (6)$$

The option would be correctly valued at V_a , and then we have:

$$dV_a - \Delta_a dS - r(V_a - \Delta_a S)dt = 0 \quad (7)$$

$$(6)-(7): dV_i - dV_a - r(V_i - V_a)dt$$

$$= e^{rt} d(e^{-rt}(V_i - V_a))$$

This is the mark-to-market profit from time t to $t + dt$.

The present value of the profit at time t_0 :

$$e^{-r(t-t_0)} e^{rt} d(e^{-rt}(V_i - V_a))$$

$$= e^{rt_0} d(e^{-rt}(V_i - V_a))$$

Then, the total profit from t_0 to maturity T is calculated by $e^{rt_0} \int_{t_0}^T d(e^{-rt}(V_i - V_a))$ (8)

Considering boundary condition: at expiration T , $V_i(T) - V_a(T) = 0$

So, for the formula (8), the final result is $V_a(0) - V_i(0)$

Conclusion: if we hedge with actual volatility, the total profit is a guaranteed amount, equal to the difference of option values, calculated by Black Scholes model with actual volatility and implied volatility respectively. However, it is a theoretical benchmark, not a trading strategy.

In order to do simulation, we use Itô-lemma to write the mark-to-market profit over time step.

Starting from formula (6): $dV_i - \Delta_a dS - r(V_i - \Delta_a S)dt$

$$dV_i = \frac{\partial V_i}{\partial t} dt + \frac{\partial V_i}{\partial S} dS + \frac{1}{2} \frac{\partial^2 V_i}{\partial S^2} dS^2$$

$$= \Theta_i dt + \Delta_i dS + \frac{1}{2} \sigma_a^2 S^2 \Gamma_i dt \quad (dS^2 = \sigma_a^2 S^2 dt) \quad (9)$$

Substitute (9) into (6): $\Theta_i dt + \Delta_i dS + \frac{1}{2} \sigma_a^2 S^2 \Gamma_i dt - \Delta_a dS - r(V_i - \Delta_a S) dt$

$$= \Theta_i dt + (\Delta_i - \Delta_a) dS + \frac{1}{2} \sigma_a^2 S^2 \Gamma_i dt - r(V_i - \Delta_a S) dt \quad (10)$$

Since, $dS = \mu S dt + \sigma_a S dX_t$,

the Black Scholes model: $\Theta_i + \frac{1}{2} \sigma_i^2 S^2 \Gamma_i + rS\Delta_i - rV_i = 0$

substitute these two formulae into (10):

$$= \frac{1}{2} (\sigma_a^2 - \sigma_i^2) S^2 \Gamma_i dt + (\Delta_i - \Delta_a) [\sigma_a S dX_t + (\mu - r) S dt]$$

Since $\mu = r$

$$= \frac{1}{2} (\sigma_a^2 - \sigma_i^2) S^2 \Gamma_i dt + (\Delta_i - \Delta_a) \sigma_a S dX_t$$

The formula shows the diffusion term, which means that how the profit is achieved is random and path-dependent.

2.1.2 Simulations of P&L

If hedging with actual volatility, the mark-to-market profit over a time step dt is

calculated with formula $\frac{1}{2} (\sigma_a^2 - \sigma_i^2) S^2 \Gamma_i dt + (\Delta_i - \Delta_a) \sigma_a S dX_t$

Based on the formula of Black Scholes, for vanilla call option, with no dividend, $D=0$:

$\Delta = N(d_1)$, where N is the cumulative distribution function of standard normal distribution

$$N(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-\frac{s^2}{2}} ds$$

$$d_1 = \frac{\ln \frac{S}{E} + (r + \frac{1}{2} \sigma^2)(T-t)}{\sigma \sqrt{T-t}}$$

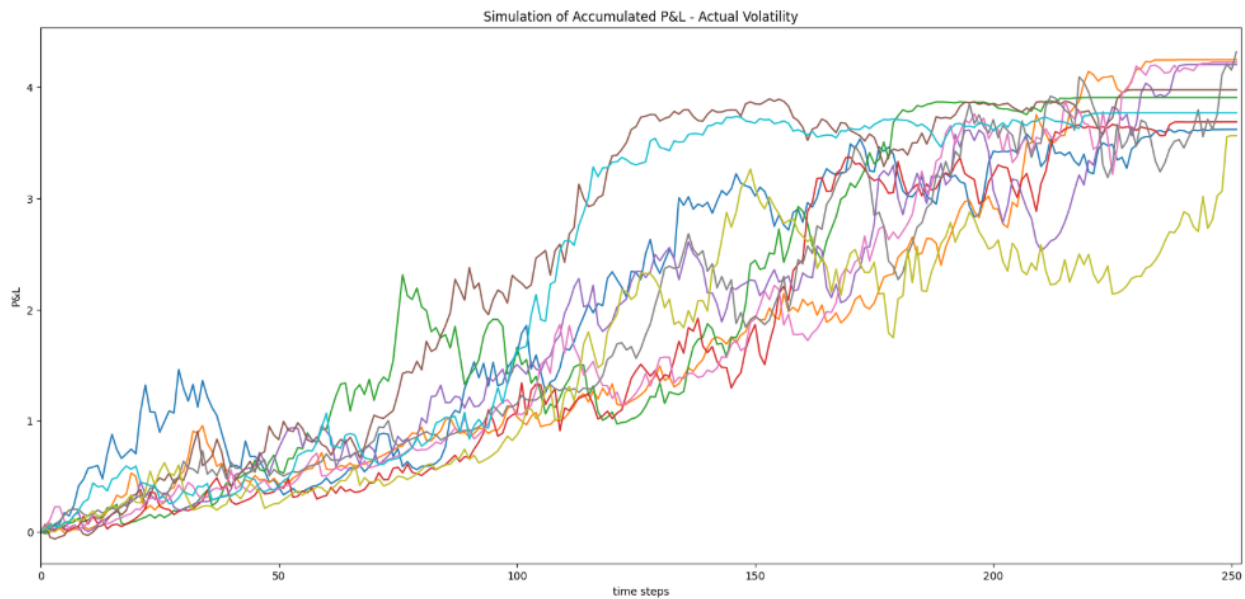
$$\Gamma = \frac{N'(d_1)}{\sigma S \sqrt{T-t}},$$

where N' is the probability density function of the standard normal distribution

$$N'(d_1) = \frac{e^{-\frac{1}{2} d_1^2}}{\sqrt{2\pi}}$$

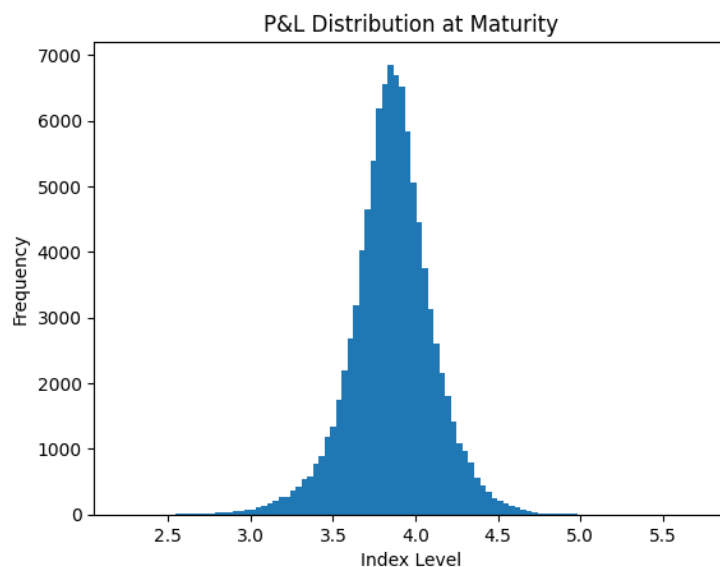
$$dX_t = \sqrt{\delta t} \phi_i, \phi_i \sim N(0,1)$$

The simulated paths show as follow:



Scenario: $S_0=100$, strike=100, rate=0.05, actual volatility= 0.3, implied volatility= 0.2, time period=1, time steps=252

We can see that the paths of the profit show random fluctuation with the implication of mark-to market risk, even though trends are upward and the final profits converge.



```
PandL[-1,:].mean(), PandL[-1,:].std()
(3.8577294305214482, 0.2515301515573105)
```

The distribution of the profit at maturity shows clear. Its mean is 3.857729, which means the expectation of final profit by delta hedging with actual volatility is 3.857729.

Also, the calculation shows that $V_a(0) - V_i(0) = 3.780671$, according to the Black Scholes formula.

This small discrepancy between simulated mean and theoretical P&L stems from discretization bias in the diffusion term.

2.2 Hedge with Implied Volatility

2.2.1 Mathematical Derivatives of the Uncertain Path-dependent Total P&L

The mark-to-market profit over time step dt is $dV_i - \Delta_i dS - r(V_i - \Delta_i S)dt$ (11)

using Itô-lemma $dV_i = \frac{\partial V_i}{\partial t} dt + \frac{\partial V_i}{\partial S} dS + \frac{1}{2} \frac{\partial^2 V_i}{\partial S^2} dS^2$

since, $dS = \mu S dt + \sigma_a S dX_t$, get $dS^2 = \sigma_a^2 S^2 dt$

so, $dV_i = \Theta_i dt + \Delta_i dS + \frac{1}{2} \sigma_a^2 S^2 \Gamma_i dt$

substitute into formula (11), get $\Theta_i dt + \frac{1}{2} \sigma_a^2 S^2 \Gamma_i dt - r(V_i - \Delta_i S)dt$ (12)

the Black Scholes formula is $\Theta_i + \frac{1}{2} \sigma_i^2 S^2 \Gamma_i + rS\Delta_i - rV_i = 0$

substitute into formula (12), and get the mark-to-market profit over time step dt is

$$\frac{1}{2} (\sigma_a^2 - \sigma_i^2) S^2 \Gamma_i dt$$

It is the deterministic time increment, a continuous-time derivation. Also, S and Γ_i change over time.

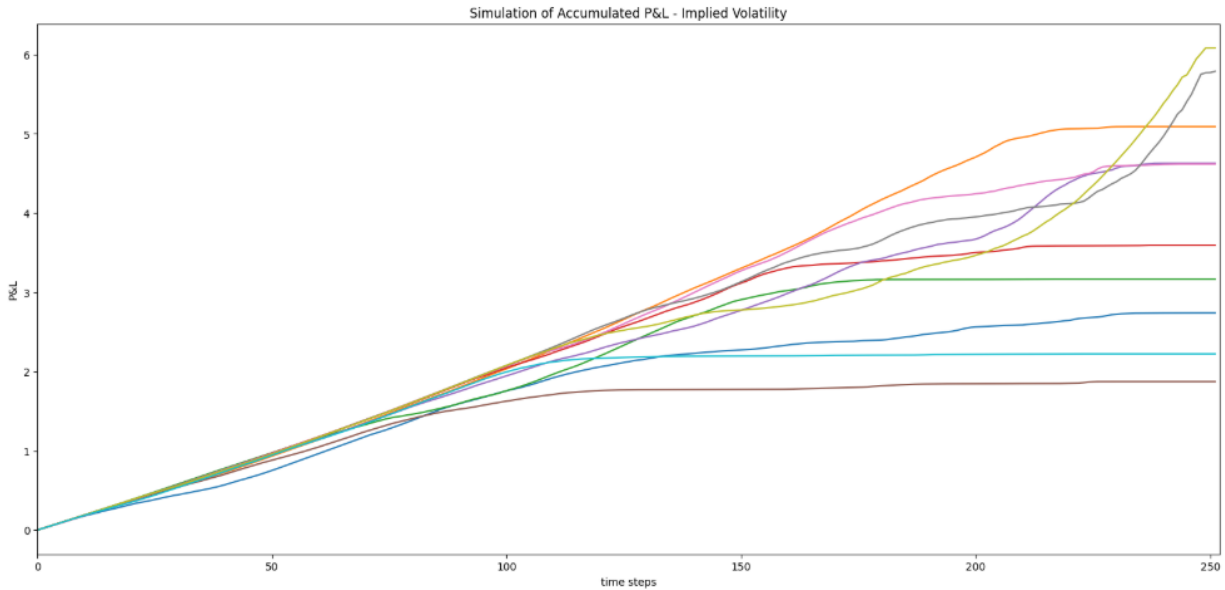
So, the accumulated profit at present time is $\frac{1}{2} (\sigma_a^2 - \sigma_i^2) \int_{t_0}^T e^{-r(t-t_0)} S^2 \Gamma_i dt$

If the forecast $\sigma_a^2 > \sigma_i^2$ is correct, the profit obtained by hedging with implied volatility is positive, with the assumption of long the option and long Gamma, $\Gamma_i > 0$. It is a model-based continuous-time result. In discrete hedging the realized P&L can deviate, since gamma, spot, hedging frequency, and realized path timing all matter.

2.2.2 Simulations of P&L

Using the mark-to-market profit over time step $dt : \frac{1}{2} (\sigma_a^2 - \sigma_i^2) S^2 \Gamma_i dt$ to do simulations.

The simulated paths show as follow:

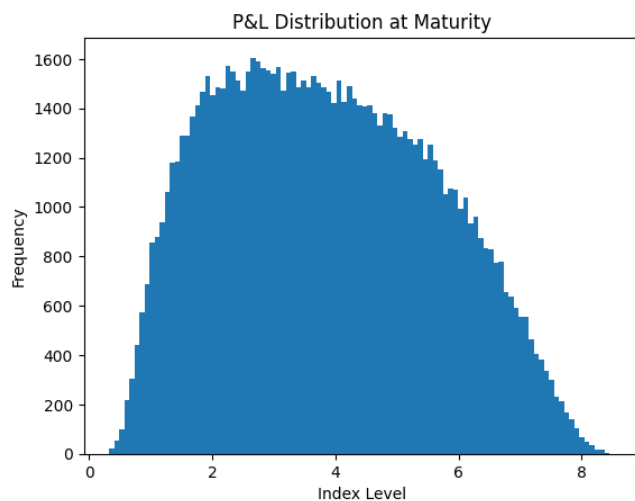


Scenario: $S_0=100$, strike=100, rate=0.05, actual volatility= 0.3, implied volatility= 0.2, time period=1, time steps=252

The simulation is the drift-dominate path under the continuous-time approximation.

Since when hedging with implied volatility, the formula of mark-to-market profit for every time step is $\frac{1}{2} (\sigma_a^2 - \sigma_i^2) S^2 \Gamma_i dt$, and it only has drift term, no diffusion term. This smooth path for accumulative P&L corresponds to the formula.

The terminal profit distribution is wider and more right-skewed.



```
PandL[-1,:].mean(), PandL[-1,:].std()
```

```
(3.8572020270436904, 1.757039616629354)
```

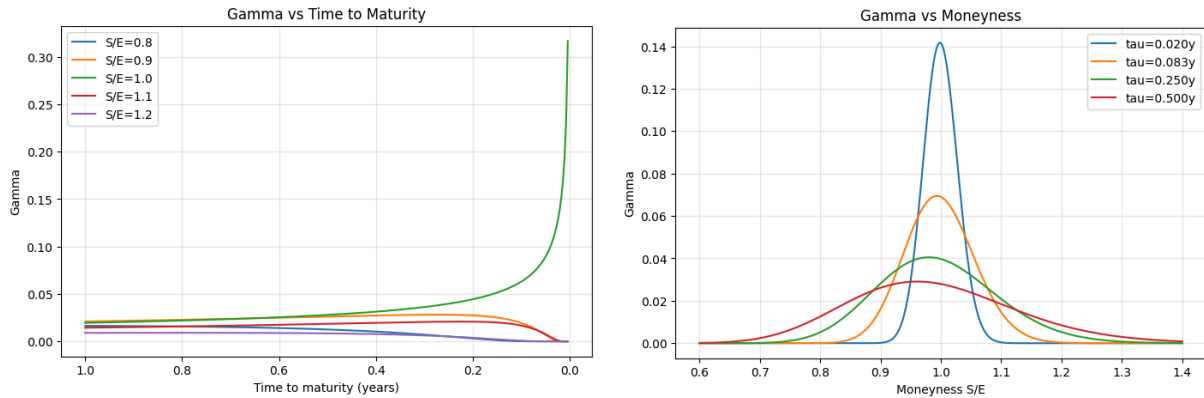
2.3 Time-dependent Gamma and Gamma Scalping Analysis

2.3.1 Gamma Analysis

The following plots show Gamma vs. Time-to-maturity for multiple moneyness and Gamma vs. Moneyness for multiple time-to-maturity.

It reflects that gamma is relatively low for long-dated options, and increases as expiry approaches. Also, gamma explosion upward happens near maturity for ATM options. Conversely, for ITM and OTM options, their gamma collapse to zero, since it becomes virtually certain whether they will finish in or out of the money.

In addition, gamma peaks ATM, and the peak becomes higher and narrower as expiry approaches. For longer time-to-maturity, the curve is wide and smooth, whereas for short-dated options, the curve shows a sharp, severe spike right at the strike price. It confirms the rule that increasing volatility (variance or time) makes gamma less severe. It also illustrates the reason that the profit of delta-hedged options depends critically on whether the spot price stays near the strike (high gamma) or drifts away.



Scenario: $S_0=100$, strike=100, rate=0.02, implied volatility= 0.2, time period=1, time steps=252

2.3.2 Gamma Scalping

Concretely, it analyzes the impact of $r^2 - \sigma_{imp}^2 dt / (\frac{dS}{S})^2 - \sigma_{imp}^2 dt$

Starting from the Itô-lemma, $dV = \frac{\partial V}{\partial t} dt + \frac{\partial V}{\partial S} dS + \frac{1}{2} \frac{\partial^2 V}{\partial S^2} dS^2$

$$dV - \Delta dS = \theta dt + \frac{1}{2} \Gamma dS^2$$

$$d(V - \Delta S) = \theta dt + \frac{1}{2} \Gamma dS^2 \quad (13)$$

Namely, P&L for delta-hedged portfolio over time step dt is approximately

$$d\pi \approx \theta dt + \frac{1}{2}\Gamma(dS)^2$$

θdt : time decay, cost of carrying the convexity

$\frac{1}{2}\Gamma(dS)^2$: gain from curvature/convexity

Based on Black Scholes formula: $\theta_i + \frac{1}{2}\sigma_i^2 S^2 \Gamma_i + rs\Delta - rV = 0$

$$\theta_i = -\frac{1}{2}\sigma_i^2 S^2 \Gamma_i - rs\Delta + rV$$

$$\theta_i = -\frac{1}{2}\sigma_i^2 S^2 \Gamma_i$$

It is the Black-Scholes theta approximation with ignorance of $-rs\Delta, rV$ terms. Here, this project ignores the small financing terms and focus on the dominant gamma-theta coupling. Because they are relatively small compared to the gamma and theta effects, especially for short-dated options or in low interest rate environments.

The formula θ_i provides connection between gamma and theta, namely convexity must be paid for through time decay (negative carry).

Substitute θ_i into the formula (13): $d(V - \Delta S) = \frac{1}{2}\Gamma dS^2 - \frac{1}{2}\sigma_i^2 S^2 \Gamma_i dt$

$$= \frac{1}{2}\Gamma[(dS)^2 - \sigma_i^2 S^2 dt]$$

$$= \frac{1}{2}\Gamma S^2 \left[\left(\frac{dS}{S}\right)^2 - \sigma_i^2 dt \right]$$

$\left(\frac{dS}{S}\right)^2 = r^2$: squared daily return/ realized variance

$$\text{So, } d(V - \Delta S) = \frac{1}{2}\Gamma S^2 [r^2 - \sigma_i^2 dt]$$

It is gamma-scalping P&L formula.

$\frac{1}{2}\Gamma S^2$ is gamma weight or dollar gamma exposure. It shows how strongly the option converts variance surprise (realized variance over the interval minus implied variance paid for over the interval) into P&L, corresponding to Jensen's convexity premium.

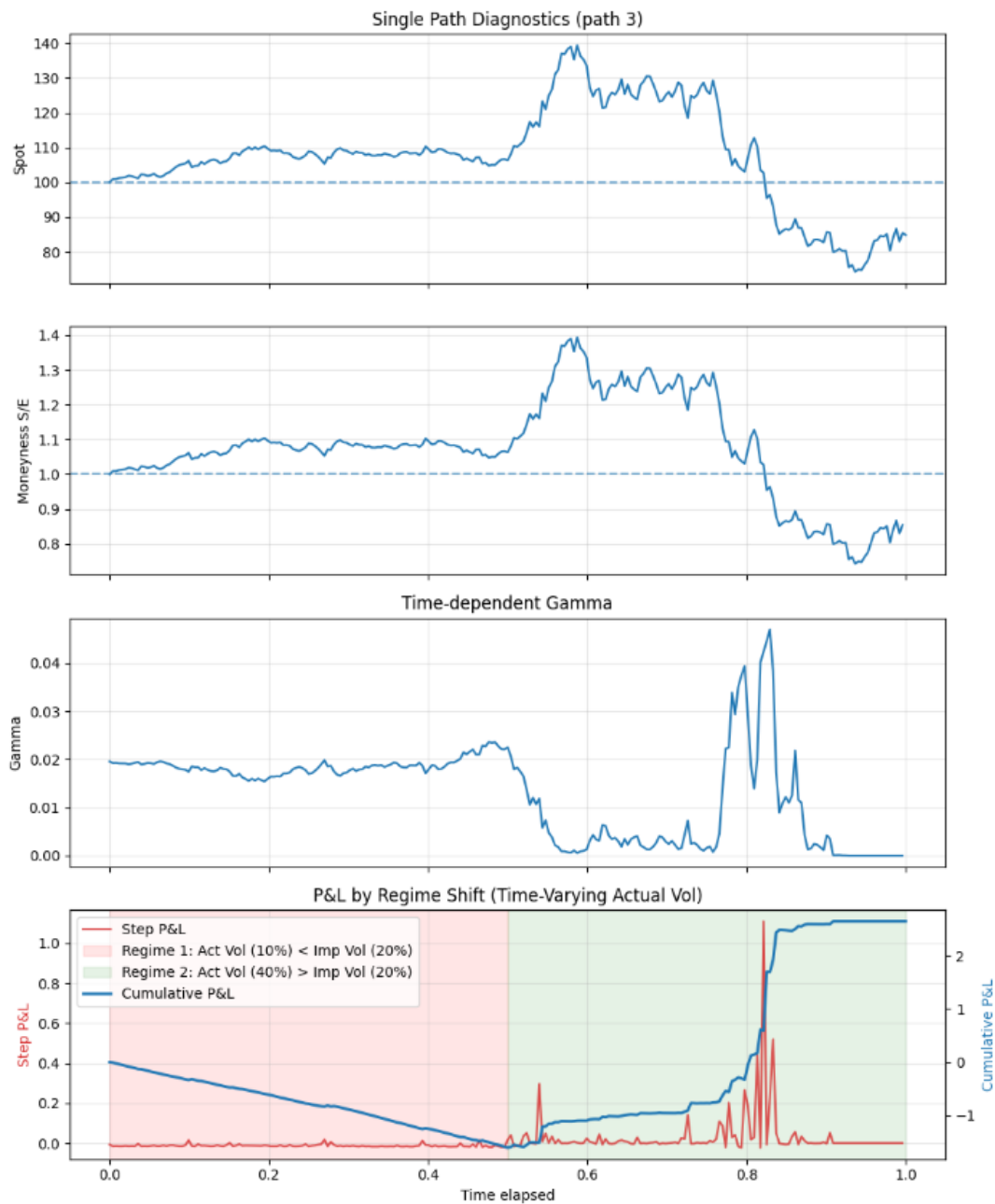
Gamma scalping (long gamma position) harvests the convexity premium through re-hedging.

The mark-to-market profit is proportional to the difference between realized and implied variance.

Accumulated P&L is $\sum_t^T \frac{1}{2}\Gamma_t S_t^2 [r_t^2 - \sigma_{t,imp}^2 dt]$

The entire analysis mirrors how volatility trading desks decompose P&L and manage Greek exposures.

The following plots show gamma scalping simulation and regime shift involved. This simulation uses discretized approximations.



Scenario: $S_0=100$, strike=100, rate=0.02, actual volatility(low)= 0.1, actual volatility(high)= 0.4, implied volatility= 0.2, time period=1, time steps=252

For regime 1 (red zone), realized volatility is 10%, it suffers a net loss, since theta decay dominates, and gamma scalping is insufficient.

For regime 1 (green zone), a regime shift occurs, and realized volatility jumps to 40%. Based on the long gamma position, delta-hedged portfolio generates profits, pushing the cumulative P&L positive.

2.3.3 Further Extensions

- 1) Apply a volatility surface (local volatility or stochastic volatility): allow various implied volatility for different strikes
- 2) Use a regime-switching model (Markov switching) to simulate the actual volatility path: a two-state Markov switching model for actual volatility would allow us to sample various regime sequences, and examine the distribution of gamma-scalping P&L conditional on regime durations and transitions.